MOHAMED WALEED M.ABDELGHANY

Software Engineer (Full-Stack, Mobile & AI)
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PROFILE

Versatile and results-driven Software Engineer with a strong foundation in Artificial Intelligence and hands-on expertise across Mobile, Full-Stack Web, and GenAI development. Adept at designing scalable architectures (Clean Architecture, SOLID) and delivering end-to-end applications from intuitive frontends to robust backend systems. Proven ability to bridge diverse technologies to consistently engineer, optimize, and deploy high-quality production-ready solutions for global digital ecosystems.

EDUCATION

Open Source Applications Development (9-Month Track) — Oct 2025 – Aug 2026
Information Technology Institute (ITI)

Bachelor of Computer and Information Sciences (Artificial Intelligence Program) — Oct 2021 – Jul 2025
Faculty of Computer and Information Sciences, Mansoura University | GPA: 3.38

PROJECTS

Mobile Development

- **Cogninue (Graduation Project):** Designed a specialized, interactive application tailored for children with autism. Engineered a comprehensive parental dashboard for monitoring, alongside a distinct child-facing interface featuring interactive games focused on improving danger cognition.
- **ModifAI:** Created an independent AI-powered photo editor in Flutter enabling text-prompt image modifications. Engineered a versatile image import system supporting device galleries, web links, shared intents, and clipboard pasting alongside Firebase integration.

Full-Stack & Web Development

- **Doctor Scheduling Management System:** Developed a comprehensive healthcare application using Vue.js for the dynamic frontend and Django REST Framework for the backend, enabling seamless appointment booking and physician availability tracking.
- **Hotel System:** Built a robust full-stack hospitality management solution utilizing Laravel, Vue.js, Inertia.js, ShadCN, and Tailwind CSS to streamline room management and client reservations.
- **Bookstore System:** Engineered an e-commerce platform utilizing Django Templates and REST Framework APIs to facilitate efficient digital inventory and secure order management.
- **Stride:** Developed a responsive Vue.js e-commerce frontend for footwear.
- **Quizzy:** Developed a Vanilla JavaScript educational platform featuring interactive quizzes and student performance tracking metrics.

AI Engineering

- **ChatGPT Clone:** Built an AI-driven chat application utilizing Hugging Face models to support dynamic text conversations while integrating text-to-image generation capabilities.
- **RAG System:** Developed a Retrieval-Augmented Generation (RAG) system utilizing LangChain, Flask, and a Vanilla JS frontend, allowing users to seamlessly upload and chat directly with PDF documents.

WORK EXPERIENCE

Software Engineer – External Communication Channels & Mobile — Nov 2026 – Present
National Bank of Egypt (NBE) | Cairo, Egypt (Full-Time)

- Engineer bank-grade mobile wallet solutions within the External Communication Channels and Mobile department.
- Collaborate closely with vendor partner Fawry to implement payment flows, digital wallet services, and secure banking integrations.

Flutter Developer Intern — Oct 2024 – Dec 2024
serv5 | Mansoura, Egypt

- Contributed to the "Green Car Wash" app by developing the UI within an MVVM architecture, integrating external APIs via Bloc, and implementing core location-based features using Firebase and Google Maps.

FREELANCE EXPERIENCE

Freelance Flutter Developer
Nov 2024 – Present

Independent

- **Mindly:** Developed a Flutter-based EdTech app offering video courses and quizzes across academic levels. Implemented advanced anti-piracy measures and a unique QR-code subscription model, successfully deploying via GitHub Actions CI/CD.
- **FarmBot:** Engineered a Flutter-based IoT agriculture dashboard displaying real-time soil and crop sensor data via REST APIs. Secured the app with Firebase Authentication to provide farmers with insights for optimizing irrigation and harvesting.
- **CX Academy:** Built a comprehensive corporate e-learning platform utilizing Flutter and GetX. Integrated real-time WebSocket chat, video streaming, quizzes, and REST APIs for seamless user and course data management.
- Delivered end-to-end cross-platform mobile and web solutions for varied clients, focusing on scalable architecture, seamless API integration, and responsive user interfaces.
- Managed complete product lifecycles independently, from initial requirement gathering and system architecture planning to final deployment on hosting platforms and app stores.

VOLUNTEERING EXPERIENCE

Head of Flutter Circle
Sep 2024 – Aug 2025

CAT Reloaded | Mansoura, Egypt

- Led a team of 55 students by developing a comprehensive Flutter curriculum and delivering high-impact technical and soft skills workshops.
- Oversaw collaborative engineering projects, guiding members on software architecture best practices to accelerate their readiness for the tech industry.

TECHNICAL SKILLS

- **Frontend Development:** React.js, Next.js, HTML, CSS, Tailwind CSS, Bootstrap, JavaScript, Vue.js
- **Backend Development:** Express.js, NestJS, PHP, Laravel, Python, Django, C++, C
- **Mobile Development:** Flutter, Dart, State Management (Provider, Bloc, Riverpod, GetX, Cubit), Push Notifications, RESTful APIs, WebSockets, MVC, MVVM, Hive
- **Database Management:** MySQL, MongoDB, Firebase Experience (Auth, Firestore, FCM), SQLite
- **Architecture & Principles:** Clean Architecture, OOP, SOLID Principles, Design Patterns
- **Tools & Deployment:** Linux/Ubuntu, Git & Version Control, CI/CD (GitHub Actions), App Store & Google Play Publishing

COURSES

- **Mobile Using Flutter (Dart, Flutter, Firebase)** – Information Technology Institute (ITI) | Jul 2024
- **Backend Development (PHP, MySQL, Laravel)** – Consultation Center of Computer and Information Science, Faculty of Computer Science, Mansoura University | Jul 2023

CERTIFICATE

- **AWS Certified Cloud Practitioner** – Amazon Web Services (AWS) (Jun 2026 - Jun 2029)
- **Mobile App Developer** – DEPI (May 2025)

MILITARY SERVICE

- **Status:** Exempted